Many protocols used in the IoT instead opt to build on top of the lower-level User Datagram Protocol (UDP), which is much simpler than TCP but provides only connectionless and unreliable delivery of messages. For example, the *Constrained Application Protocol* (CoAP), provides an alternative to HTTP for constrained devices and is based on UDP. To protect these protocols, a variation of TLS known as Datagram TLS (DTLS) has been developed.[1]

> **DEFINITION** ***Datagram Transport Layer Security*** (DTLS) is a version of TLS designed to work with connectionless UDP-based protocols rather than TCP-based ones. It provides the same protections as TLS, except that packets may be reordered or replayed without detection.

Recent DTLS versions correspond to TLS versions; for example, DTLS 1.2 corresponds to TLS 1.2 and supports similar cipher suites and extensions. At the time of writing, DTLS 1.3 is just being finalized, which corresponds to the recently standardized TLS 1.3.

QUIC

A middle ground between TCP and UDP is provided by Google's QUIC protocol (Quick UDP Internet Connections, https://en.wikipedia.org/wiki/QUIC), which will form the basis of the next version of HTTP: HTTP/3. QUIC layers on top of UDP but provides many of the same reliability and congestion control features as TCP. A key feature of QUIC is that it integrates TLS 1.3 directly into the transport protocol, reducing the overhead of the TLS handshake and ensuring that low-level protocol features also benefit from security protections. Google has already deployed QUIC into production, and around 7% of Internet traffic now uses the protocol.

QUIC was originally designed to accelerate Google's traditional web server HTTPS traffic, so compact code size was not a primary objective. But the protocol can offer significant advantages to IoT devices in terms of reduced network usage and low-latency connections.